[redacted] Chat Log

Transformation Code

```
SELECT
    'AGE' AS variable_name,
    AGE AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
    'SMOKING' AS variable_name,
    SMOKING AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
    'ANXIETY' AS variable_name,
    ANXIETY AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
    'CHRONIC_DISEASE' AS variable_name,
    CHRONIC_DISEASE AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
    'ALCOHOL_CONSUMING' AS variable_name,
    ALCOHOL_CONSUMING AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
    'COUGHING' AS variable_name,
    COUGHING AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
    'SHORTNESS_OF_BREATH' AS variable_name,
    SHORTNESS_OF_BREATH AS value,
    GENDER
```

```
FROM dataset
UNION ALL
SELECT
    'CHEST_PAIN' AS variable_name,
    CHEST_PAIN AS value,
    GENDER
FROM dataset
```

Derivation Dialog

user

[CONTEXT]

[TABLE dataset]

Column metadata:

GENDER (VARCHAR) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ["('F',)", "('M',)"]}
AGE (BIGINT) ---- {'count': 3000, 'unique_count': 51, 'null_count': 0, 'sample_values': ['(41,)', '(46,)', '(56,)', '(30,)', '(65,)', '(67,)', '(62,)']}
SMOKING (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
YELLOW_FINGERS (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
ANXIETY (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
PEER_PRESSURE (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
CHRONIC_DISEASE (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
FATIGUE (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
ALLERGY (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
WHEEZING (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
ALCOHOL_CONSUMING (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
COUGHING (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
SHORTNESS_OF_BREATH (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
SWALLOWING_DIFFICULTY (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}
CHEST_PAIN (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}

LUNG_CANCER (VARCHAR) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ["('YES',)", "('NO',)"]}

Sample data:

| GENDER | AGE | SMOKING | YELLOW_FINGERS | ANXIETY | PEER_PRESSURE | CHRONIC_DISEASE | FATIGUE | ALLERGY | WHEEZING | ALCOHOL_CONSUMING | COUGHING | SHORTNESS_OF_BREATH | SWALLOWING_DIFFICULTY | CHEST_PAIN | LUNG_CANCER |
0| M | 65 | 1 | 1 | 1 | 2 | 2 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | NO |
1| F | 55 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | NO |
2| F | 78 | 2 | 2 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 1 | 2 | 1 | 1 | YES |
3| M | 60 | 2 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 1 | 2 | 1 | 2 | 2 | YES |
4| F | 80 | 1 | 1 | 2 | 1 | 1 | 2 | 1 | 2 | 1 | 1 | 1 | 1 | 2 | NO |

[GOAL]

What are the main characteristics of the dataset?

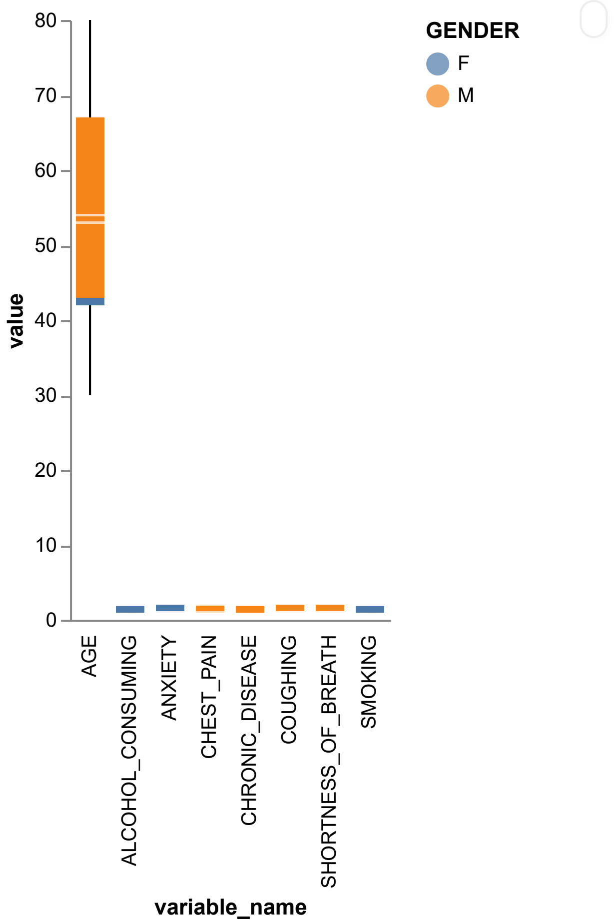
[OUTPUT]

assistant

```json
{
  "mode": "overview",
  "recommendation": "To understand the main characteristics of the dataset, we need to provide an overview of the distribution of key variables. The dataset contains information about patient demographics and health conditions. We will reshape the data from wide format to long format to make it more suitable for analysis and visualization.",
  "output_fields": ["variable_name", "value", "GENDER"],
  "chart_type": "boxplot",
  "visualization_fields": ["variable_name", "value", "GENDER"]
}
```

```sql
SELECT
    'AGE' AS variable_name,
    AGE AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
    'SMOKING' AS variable_name,
    SMOKING AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
    'ANXIETY' AS variable_name,
    ANXIETY AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
    'CHRONIC_DISEASE' AS variable_name,
    CHRONIC_DISEASE AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
    'ALCOHOL_CONSUMING' AS variable_name,
    ALCOHOL_CONSUMING AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
    'COUGHING' AS variable_name,
    COUGHING AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
```

```
    'SHORTNESS_OF_BREATH' AS variable_name,
    SHORTNESS_OF_BREATH AS value,
    GENDER
FROM dataset
UNION ALL
SELECT
    'CHEST_PAIN' AS variable_name,
    CHEST_PAIN AS value,
    GENDER
FROM dataset
```



AI generated results can be inaccurate, inspect it!

sample data from **view_bxek** (3 columns × 24000 rows)

show column stats

variable_name	value	GENDER
AGE	65	M
AGE	55	F
AGE	78	F
AGE	60	M
AGE	80	F
AGE	58	F
AGE	70	F
AGE	74	F
AGE	77	M
......	......	......